Identification

Term	Definition
1. Data Flow Diagram	Takes an input-process-output view. Data objects flow into software, are transformed by processing elements, and resultant data objects flow out.
2. Cost	Refers to the estimated resources and expenses required to complete a software project.
3. Waterfall Model	Sometimes called the *classic life cycle* suggests a systematic, sequential approach beginning with customer specification of requirements and progressing through planning, modeling, construction, and deployment.
4. Context Diagram	Also known as *Level 0 DFD* is the first and highest-level Data Flow Diagram that represents the entire system as a whole.
5. Requirement	May range from a high-level abstract statement of a service to a detailed mathematical functional specification.
6. Effort	Refers to the amount of work required to complete a software project.
7. Process	Defines the framework that holds the technology layers together; Is a collection of work activities, actions, and tasks performed when some work product is to be created.
8. System Requirements	A structured document setting out detailed descriptions of the system's functions, services, and operational constraints. Defines what should be implemented; may be part of a contract.
9. Software Project Management	Refers to the branch of project management dedicated to planning, scheduling, resource allocation, execution, tracking, and delivery of software and web projects.
10. Quality Focus	The bedrock commitment to quality underlies all software engineering practice.
11. Risk Categorization	Is the process of classifying identified project risks by type and impact as part of the overall Risk Management Process.
12. Effort Estimation	Is the process of determining the amount of work required to complete a software project.

13. Planning	Defines the engineering work, resources, risks, and schedule for a software project; the activity that isolates requirements, develops size and resource estimates, identifies development tasks, and creates a project schedule.
14. Software Reengineering	Encompasses inventory analysis, document restructuring, reverse engineering, program and data restructuring, and forward engineering.
15. System Software	Collection of programs written to service other programs.
16. Spiral Model	*Barry Boehm (1988)* couples the iterative nature of prototyping with the controlled, systematic aspects of the waterfall model. Provides rapid development of increasingly complete software versions. Key feature: explicit risk assessment at each iteration loop.
17. Agile	Agile methods use User Stories instead of formal requirements documents (XP approach); Uses incremental requirements engineering; Is practical for business systems but problematic for critical systems or systems developed by several teams.
18. Software Process	Is a collection of work activities, actions, and tasks that are performed when some work product is to be created, where each activity, action, and task resides within a framework or model that defines their relationship with the process and with one another.
19. Embedded Software	Resides within a product or system.
20. Use Case Diagram	Step-by-step stories of how users perform tasks.
21. V-Model	A variation of the Waterfall model. Depicts the relationship of QA actions to communication, modeling, and early construction activities.
22. Milestone	Is a key checkpoint used to indicate and measure progress within a software project; verify key deliverables are achieved.
23. Technical Review	Is an umbrella activity performed across all software engineering process activities to ensure and maintain the quality of work products throughout the software development process.
24. Scope Management	Is the process of defining, verifying, and controlling the boundaries of a software project.

25. Functional Requirement	Statements of services the system should provide, how it should react to inputs, and how it should behave in situations. May state what the system should NOT do.
26. Requirements Specification	Is the process of writing down the user and system requirements in a requirements document
27. Software Engineering	is an engineering approach on software development involving systematic application of analyzing user needs and designing, constructing, and testing end-user applications.
28. Artificial Intelligence Software	Makes use of non-numerical algorithms to solve complex problems.
29. Application Software	Stand-alone programs that solve a specific business need.
30. Non-Functional Requirement	Constraints on the services or functions offered by the system (timing constraints, development process constraints, standards). Often apply to the system as a whole.
31. Software Maintenance	Begins almost immediately after software is released to end users. Corrects defects, adapts software to changing environments, and enhances functionality to meet evolving needs.
32. Incremental Model	Combines elements of linear and parallel process flows. Applies linear sequences in a staggered fashion. Each sequence produces deliverable 'increments' of the software.